300490

Course Name: Introduction to Microprocessor

Course Outcome

CO1- Demonstrate the Microprocessor internal architecture, instructions and its operations
CO2- Develop programs based on 8085 microprocessor instruction set and addressing mode
CO3- Develop program using looping, counting, indexing, counter and time delays.
CO4- Compare accepted standards and guidelines to select microprocessor (8085 & 8086) to meet performance requirement
CO5- Analyze the concept of interfacing the processor to external device with I/O programming and interrupt driven I/O employing different address space mapping.
CO6- Understand the working of general-purpose devices (8237, 8254, 8255 & 8259)

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End Term Examination, Even Semester 2024-25
B.Tech. (CSE/AIML & IIoT/BCT/CCV/DA/CSF), Year II, Semester IV
BCSC 0007: Introduction to Microprocessor

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. All parts of a question should be answered at one place.
2. First, complete the one set of section then start a new section.
3. Assume the appropriate data for any missing data.
4. Use suitable diagrams to explain the answer.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Compare the following Intel's 8085 instructions. (i) LDA and STA (ii) MOV and MVI (iii) RAL and RLC (iv) CMP and SUB	4	1	An	F
2	List the names of different arithmetic and logical operations performed by the ALU of 8085 µP. Determine the value of Accumulator and Status flag register (in Hex) for 8085 Microprocessor after executing the following sequence of instructions. LXI H, 2400 H MOV A, M INX H ADD M	4	2	U	C

	HLT Assume [2400] = 4D H and [2401] = A3 H				
3	Draw the pin diagram of 8085 microprocessor and explain the functions of different signals of timing and control unit.	4	1	U	P
4	Count the number of even numbers of a series by writing an assembly language program for 8085 Microprocessor. The length of the series is stored in the memory location 24FF H and the series begins from 2500 H. Store the count of even number at the memory location 3500 H.	4	3	A	P
5	Suppose you're building a computer system for a unique project, and your team is thinking about using the 8086 Microprocessor. Describe how the 8086's design, including its main parts and pipelining ability, makes it better than the 8085 Microprocessor for this task. Or Imagine you are designing a system for a high-speed data acquisition application, where data from multiple sensors needs to be continuously transferred to memory for processing. Describe how you would utilize 8237 DMA controller to efficiently handle these data transfers.	4	4,6	An, U	C, P

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Write a BSR control word subroutine to set bits PC_7 & PC_3 and reset them after 15 ms. Assume a delay subroutine is available. Let the address of the control word register (CWR) is 83 H.	3	6	U	F
7	Differentiate between I/O mapped I/O and memory mapped I/O schemes of address mapping for I/O devices. Compare the advantages and disadvantages of these schemes.	3	5	An	C
8	Explain Intel 8254 Programmable Interval Timer (PI T) with suitable block diagram.	3	3	R	F
9	Draw the conditional and control bit pattern for the	3	4	U	C

	flag register of 8086 and discuss the significance of each bit.				
10	In a home security system, an Intel 8255 PPI is used to interface a microprocessor with I/O devices. All its three ports (Port A, Port B & Port C) are connected with different sensors to estimate and control certain physical parameters. Explain the functional block diagram of the 8255 PPI and how its components enable the microprocessor to communicate with the external devices.	3	5	A	F

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Write a program for 8085 Microprocessor to count from '0' to '9' with one second of delay between each count. At the count of 9, the counter should reset itself to '0' and repeat the sequence continuously. Use register pair H-L to setup the delay, and display each count at one of the output ports. Assume the clock period of the microcomputer is 1 μsec.	5	3	C	P
12	Explain different types of addressing modes of 8086 Microprocessor with suitable examples. Obtain the 20-bit physical address generated by the BIU if the contents of CS = 134C H and IP = 241D H in the 8086 Microprocessor.	5	4	A	C
13	Imagine you're a hardware engineer designing an interrupt management system for a computer setup using the Intel 8085 microprocessor, and you've chosen to integrate the Intel 8259 Programmable Interrupt Controller (PIC) to handle multiple peripheral interrupts. Explain what the Intel 8259 PIC is, and describe its architecture, highlighting the roles of its different registers in managing interrupts for this system.	5	6	R	P